1. A random variable X has a mean 8 and variable 9 and an unknown probability distribution. Find $P(-4 < X < 20)$ and $P(|X-8| \geq 6)$
2. A random variable X has a mean 12 and variance of 9 and an unknown probability distribution. Find $P(6 < X < 18)$ and $P(3 < X < 21)$
3. A random variable has a probability distribution

x	0	1	2	3
p(x)	1/2	1/4	1/8	1/8

   i. Find the upper bound for $P(|X-1| > 2)$ by Chebychev's inequality.

   ii. Find $P(|X-1| > 2)$ by direct computation.

4. A random variable has a density function e^{-x}, $x \geq 0$. Find the upper limit of $P(|X-1| > 2)$
5. A random variable X has mean 1o and variance 4 and an unknown probability distribution. Find the value of C such that $P(|X-10| \geq C) \leq 0.04$

Some questions for practices

1. Evaluate E[Y/X]; $f(x,y) = \begin{cases} \dfrac{x(1+3y^2)}{4}, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{else} \end{cases}$

2. The pdf of the RV X is given $f(x) = \begin{cases} \dfrac{1}{4}, & \text{-2} < x < 2 \\ 0, & \text{else} \end{cases}$

   Find (i) $P(|X| > 1)$, (ii) $P(X > 1.5 \mid X > 1)$

3. Cumulative function is given, $F(x) = \begin{cases} 0 & , \quad x < 0 \\ \dfrac{x^2}{16} & , \quad 0 \leq x \leq 4 \\ 1 & , \quad 4 \leq x \end{cases}$

   Find $P(X > 1 \mid X < 3)$